

Caifen Liu

PhD Candidate in Epidemiology and Biostatistics

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EDUCATION

The University of Hong Kong

PhD in Epidemiology and Biostatistics

Thesis: *Estimation and Interpretation of COVID-19 Severity and Vaccine Effectiveness.*

Supervisor: [Prof. Ben Cowling](#).

Hong Kong SAR, China

Sep 2022–Sep 2026 (expected)

Southern University of Science and Technology

MSc in Statistics

Thesis: *Bivariate Binomial Distribution for Comparing Binomial Observations in Two Correlated Groups.*

Supervisor: [Prof. Guo-Liang Tian](#).

Shenzhen, China

Sep 2018–Jun 2020

Hunan University

BE in Materials Science and Engineering

Changsha, China

Sep 2014–Jun 2018

RESEARCH INTERESTS

I am interested in developing and applying causal inference methods to observational health data to reduce bias, strengthen the validity of causal conclusions, and support clinical decision-making.

Methodology: causal inference; causal estimands; selection bias; inverse-probability weighting; survival and longitudinal data analysis; simulation and sensitivity analyses; missing data methods; target trial emulation.

Applications: observational studies using linked electronic health records, registries, and cohort data; vaccine effectiveness evaluation; disease severity evaluation; pharmacoepidemiology.

DOCTORAL RESEARCH

The University of Hong Kong *School of Public Health*

2022–Present

COVID-19 severity pyramid and in-hospital progression

Investigated how changing admission practices altered COVID-19 severity estimates and longitudinal in-hospital progression. Modeled about 2.8 million linked case records, including about 120,000 hospitalizations, using Bayesian severity-pyramid, Aalen–Johansen competing-risk, and Cox models.

Hospital pressure, admission selection, and fatality risk

Assessed how hospital pressure, admission selection, and admission severity shaped the observed in-hospital fatality. Applied period-specific generalized additive models and model-based counterfactual standardization to about 120,000 hospitalizations.

Causal estimands and selection bias in test-negative designs

Examined when vaccine-induced severity attenuation drives test-negative estimates away from source-population causal estimands. Used causal DAGs, analytic bias derivations, Monte Carlo simulation, and inverse-probability weighting to evaluate bias and correction strategies.

Hospital-based test-negative design estimates of vaccine effectiveness (*ongoing*)

Uses simulation to evaluate when hospital-based TNDs recover vaccine effectiveness against target-pathogen hospitalization, benchmarked against randomized-trial and adjusted-cohort estimates under alternative hospital eligibility, testing, and control-selection mechanisms.

Peer-Reviewed Publications

†Co-first author

1. **Liu, C.**, Okoli, G. N., Chen, R., Sullivan, S. G., and Cowling, B. J. (2026). SARS-CoV-2 vaccination and attenuation of breakthrough infection severity: A systematic global review and meta-analysis. *Clinical Infectious Diseases*, ciag346.
2. Du, Z., **Liu, C.**[†], Bai, Y., Wang, L., Lim, W. W., Lau, E. H. Y., and Cowling, B. J. (2024). Predicting efficacies of fractional doses of vaccines by using neutralizing antibody levels. *JMIR Public Health and Surveillance*, 10, e49812.
3. Du, Z., **Liu, C.**[†], Wang, C., Xu, L., Xu, M., Wang, L., Bai, Y., Xu, X., Lau, E. H. Y., Wu, P., and Cowling, B. J. (2022). Reproduction numbers of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants: A systematic review and meta-analysis. *Clinical Infectious Diseases*, 75(1), e293–e295.
4. Du, Z., Wang, C., **Liu, C.**[†], Bai, Y., Pei, S., Adam, D. C., Wang, L., Wu, P., Lau, E. H. Y., and Cowling, B. J. (2022). Systematic review and meta-analyses of superspreading of SARS-CoV-2 infections. *Transboundary and Emerging Diseases*, 69(5).
5. **Liu, C.**[†], Xu, L., Bai, Y., Xu, X., Lau, E. H. Y., Cowling, B. J., and Du, Z. (2022). Local surveillance of the COVID-19 outbreak. *Frontiers in Physics*, 10, 824369.

Working Papers

1. **Liu, C.**, Murphy, C., Okoli, G. N., Sullivan, S. G., and Cowling, B. J. *Vaccine-induced severity attenuation and bias in test-negative estimates of vaccine effectiveness*. Manuscript in preparation.
2. **Liu, C.**, Cheung, J. K., Lin, Y., Wu, P., Wong, J. Y., and Cowling, B. J. *Hospital pressure, admission selection, and COVID-19 hospital fatality risk across epidemic periods in Hong Kong: a retrospective cohort study*. Manuscript in preparation.

SELECTED PRESENTATIONS

EPIDEMICS¹⁰

San Diego, USA

Oral presentation

Dec 2025

Impact of vaccine-induced severity attenuation on bias in vaccine-effectiveness estimates from test-negative designs.

EPIDEMICS⁹

Bologna, Italy

Poster presentation

Nov 2023

Estimating the COVID-19 severity pyramid in Hong Kong, January 2020–January 2023.

RESEARCH EXPERIENCE

Laboratory of Data Discovery for Health (D²4H)

Hong Kong SAR, China

Research Associate

Jul 2021–Jul 2022

- Supported data access, protocol development, and ethics approval for a Hospital Authority Data Collaboration Lab retrospective cohort study of oseltamivir effectiveness in hospitalized influenza patients using longitudinal clinical records.
- Co-developed an R package integrating EpiEstim and virosolver to simulate epidemic trajectories, estimate time-varying reproduction numbers, and support real-time monitoring of local COVID-19 transmission.

- Conducted systematic reviews and meta-analyses of SARS-CoV-2 transmissibility, synthesizing variant-specific reproduction numbers and superspreading parameters.

Southern University of Science and Technology
Research Assistant, Department of Statistics and Data Science

Shenzhen, China
Jul 2020–Jun 2021

- Conducted methodological literature reviews and developed estimation methods using the MM algorithm for bivariate binomial distributions and multivariate zero-inflated count data.

TEACHING EXPERIENCE

The University of Hong Kong
CMED 6020 Advanced Statistical Methods I, *Tutor*
CMED 6020 Advanced Statistical Methods I, *Teaching Assistant*

Spring 2025–Spring 2026
Spring 2023–Spring 2024

Southern University of Science and Technology
MA103A Linear Algebra A, *Teaching Assistant*
MA102B Calculus A II, *Teaching Assistant*
MA101B Calculus A I, *Teaching Assistant*

Fall 2019
Spring 2019
Fall 2018

ADDITIONAL TRAINING

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)
University of Washington, Department of Biostatistics

July 2023

Selected modules: MCMC I and II for Infectious Diseases; Simulation-Based Inference for Epidemiological Dynamics; Statistics and Modeling with Novel Data Streams.

TECHNICAL SKILLS

Programming: R, Python; **Tools:** $\LaTeX$, Git.

LANGUAGES

Mandarin (native); English (professional); Cantonese (conversational).